Ajeet Kumar

Pursuing B.Tech in Computer Science and Engineering (AI & ML)

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PROFILE

B.Tech CSE (AI & ML) student with strong fundamentals in Python, Machine Learning Algorithms, and Data Processing, and hands-on industry experience building an AI-powered, full-stack retail analytics platform. Skilled in NLP/GenAI, AI Model Development, computer vision, and Cloud/ML Frameworks, with a track record of shipping production-style projects end to end. Seeking AI Engineer roles; open to relocating to Coimbatore.

EDUCATION

- **Ghani Khan Choudhury Institute of Engineering and Technology (GKCIET)** 2023 – 2027
- B.Tech in Computer Science and Engineering (Artificial Intelligence & Machine Learning) | Affiliated to MAKAUT

EXPERIENCE

Artificial Intelligence Intern — Infosys Springboard

Infosys Springboard Virtual Internship 7.0 — AI-Powered Consumer Attention Mapping System GitHub

Remote | July 2026 – October 2026

- Architected Retail Analytics System: Designed and deployed an end-to-end AI consumer attention mapping platform using FastAPI microservices and Docker (Cloud/ML deployment) to analyze shopper engagement and store traffic patterns in real-time.
- Built Computer Vision & AI Model Pipelines: Integrated YOLOv8, ByteTrack, and MediaPipe to perform multi-person tracking across store zones, re-identification (Re-ID), and gaze/head-pose estimation for dwell-time calculation.
- Engineered Composite Attractiveness Engine: Developed an ML algorithm combining dwell duration, pickup rates, and conversion frequency to generate real-time product visibility and shelf optimization scores.
- Implemented Multi-Tier Data Processing Architecture: Structured high-throughput data workflows using PostgreSQL for relational data, TimescaleDB for time-series dwell logs, and Redis for caching active tracking sessions.
- Delivered Role-Based Intelligence Dashboards: Created interactive web interfaces using React.js and dynamic heatmap visualizations to provide actionable merchandising insights for store managers and retail analysts.

PROJECTS

Your AI Partner — Full-Stack AI EdTech Platform (GitHub | Live Demo)

- Developed a dark-themed glassmorphic EdTech dashboard featuring a personalized AI Doubt Solver, adaptive Mock Exam engine (NLP/GenAI), and 3D Revision Flashcards.
- Built a hybrid AI pipeline combining local private inference (Ollama) and high-speed cloud computation (Google Gemini 2.5 Flash) — applied AI Model Development and Cloud/ML Frameworks.
- Implemented multimodal vision input allowing users to upload photo queries of equations/textbooks for step-by-step AI breakdowns.
- Deployed the full stack using Vercel serverless hosting and engineered a custom self-healing Cloudflare tunnel manager daemon for local API synchronization.

XAI-BioSign: Explainable Ensemble Framework for Multi-Omics Biomarker Discovery (In Progress)

- Developing an explainable AI framework that applies ensemble learning (ML Algorithms) on multi-omics datasets to identify interpretable biomarkers for disease classification.

Medical Image Captioning using Deep Learning (GitHub)

- Developed a hybrid deep learning model combining CNNs and LSTMs in TensorFlow/Keras to automatically generate diagnostic text captions for radiology images.
- Processed a large-scale dataset of 73,000+ image-caption pairs (ROCO dataset), utilizing OpenCV for image normalization and NLTK for text tokenization.
- Engineered a visual feature extraction pipeline using transfer learning with pre-trained VGG16 and ResNet50 models to capture complex medical imaging features.
- Evaluated caption generation performance using quantitative NLP metrics (BLEU, ROUGE, METEOR), optimizing the decoder for accurate medical terminology.

SKILLS

- **Languages:** C, Java, Python
- **AI / ML & NLP/GenAI:** YOLOv8, ByteTrack, DeepSORT, OpenCV, MediaPipe, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn, Streamlit, Google Gemini API, Ollama
- **Cloud & ML Frameworks:** Docker, Vercel, Google Gemini API, Ollama, PyTorch, TensorFlow
- **Backend, Systems & Data Processing:** FastAPI, REST APIs, Microservices, Docker, Kafka, Redis, JWT/RBAC, PostgreSQL, TimescaleDB, MongoDB
- **Frontend & Visualization:** React.js, Tailwind CSS, Plotly, Chart.js, MVC, Swing
- **Tools & Platforms:** Git, Vercel
- **Soft Skills:** Problem-solving, Analytical thinking, Technical writing, Fast learner

REFERENCES

- Dr. Sukhen Das Mondal, Asst. Professor, GKCIET — sukhen@gkciet.ac.in
- Dr. Showmik Bhowmik, Asst. Professor, GKCIET — showmik@gkciet.ac.in

DECLARATION

I hereby declare that the information provided above is accurate and truthful.